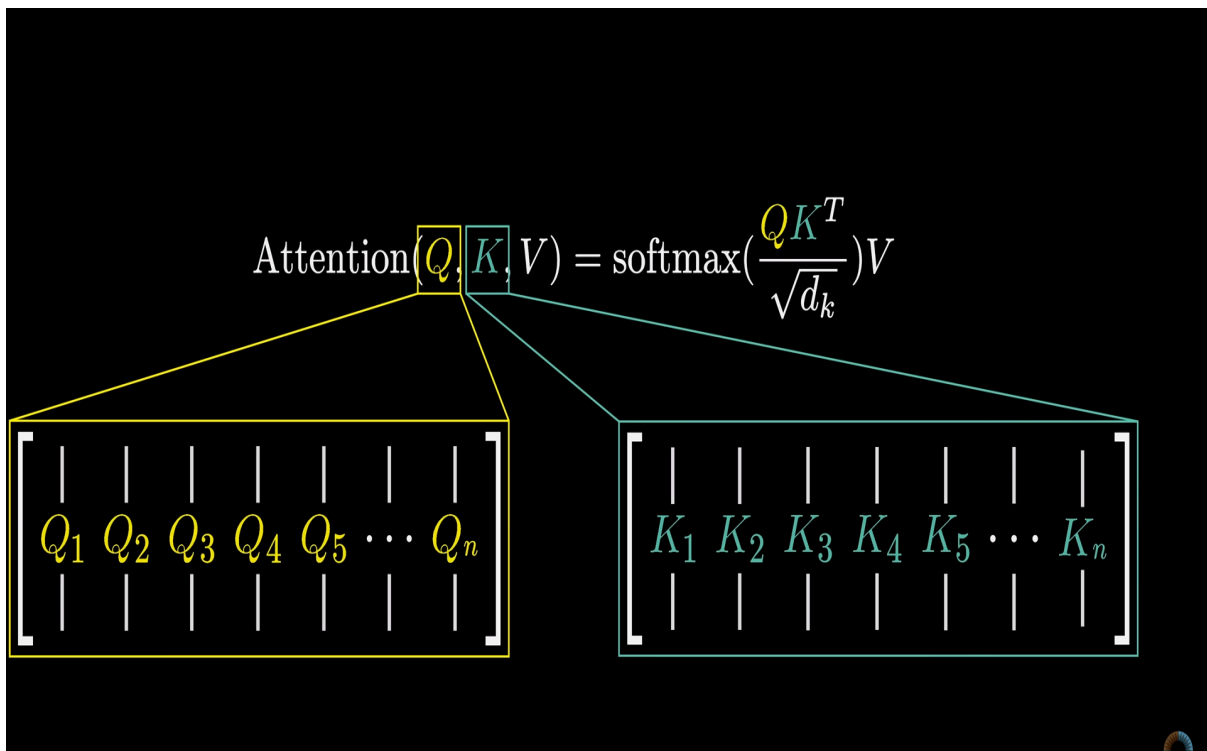


Transformers

	$\begin{array}{c} \boxed{\text{a}} \\ \downarrow \\ \vec{E}_1 \\ \downarrow^{W_Q} \\ \vec{Q}_1 \end{array}$	$\begin{array}{c} \boxed{\text{fluffy}} \\ \downarrow \\ \vec{E}_2 \\ \downarrow^{W_Q} \\ \vec{Q}_2 \end{array}$	$\begin{array}{c} \boxed{\text{blue}} \\ \downarrow \\ \vec{E}_3 \\ \downarrow^{W_Q} \\ \vec{Q}_3 \end{array}$	$\begin{array}{c} \text{creature} \\ \downarrow \\ \vec{E}_4 \\ \downarrow^{W_Q} \\ \vec{Q}_4 \end{array}$	$\begin{array}{c} \boxed{\text{roamed}} \\ \downarrow \\ \vec{E}_5 \\ \downarrow^{W_Q} \\ \vec{Q}_5 \end{array}$	$\begin{array}{c} \boxed{\text{the}} \\ \downarrow \\ \vec{E}_6 \\ \downarrow^{W_Q} \\ \vec{Q}_6 \end{array}$	$\begin{array}{c} \boxed{\text{verdant}} \\ \downarrow \\ \vec{E}_7 \\ \downarrow^{W_Q} \\ \vec{Q}_7 \end{array}$	$\begin{array}{c} \boxed{\text{forest}} \\ \downarrow \\ \vec{E}_8 \\ \downarrow^{W_Q} \\ \vec{Q}_8 \end{array}$	
$\boxed{\text{a}} \rightarrow \vec{E}_1 \xrightarrow{W_k} \vec{K}_1$	$\vec{K}_1 \cdot \vec{Q}_1$	$\vec{K}_1 \cdot \vec{Q}_2$	$\vec{K}_1 \cdot \vec{Q}_3$	$\vec{K}_1 \cdot \vec{Q}_4$	$\vec{K}_1 \cdot \vec{Q}_5$	$\vec{K}_1 \cdot \vec{Q}_6$	$\vec{K}_1 \cdot \vec{Q}_7$	$\vec{K}_1 \cdot \vec{Q}_8$	
$\boxed{\text{fluffy}} \rightarrow \vec{E}_2 \xrightarrow{W_k} \vec{K}_2$	$\vec{K}_2 \cdot \vec{Q}_1$	$\vec{K}_2 \cdot \vec{Q}_2$	$\vec{K}_2 \cdot \vec{Q}_3$	$\vec{K}_2 \cdot \vec{Q}_4$	$\vec{K}_2 \cdot \vec{Q}_5$	$\vec{K}_2 \cdot \vec{Q}_6$	$\vec{K}_2 \cdot \vec{Q}_7$	$\vec{K}_2 \cdot \vec{Q}_8$	
$\boxed{\text{blue}} \rightarrow \vec{E}_3 \xrightarrow{W_k} \vec{K}_3$	$\vec{K}_3 \cdot \vec{Q}_1$	$\vec{K}_3 \cdot \vec{Q}_2$	$\vec{K}_3 \cdot \vec{Q}_3$	$\vec{K}_3 \cdot \vec{Q}_4$	$\vec{K}_3 \cdot \vec{Q}_5$	$\vec{K}_3 \cdot \vec{Q}_6$	$\vec{K}_3 \cdot \vec{Q}_7$	$\vec{K}_3 \cdot \vec{Q}_8$	
$\boxed{\text{creature}} \rightarrow \vec{E}_4 \xrightarrow{W_k} \vec{K}_4$	$\vec{K}_4 \cdot \vec{Q}_1$	$\vec{K}_4 \cdot \vec{Q}_2$	$\vec{K}_4 \cdot \vec{Q}_3$	$\vec{K}_4 \cdot \vec{Q}_4$	$\vec{K}_4 \cdot \vec{Q}_5$	$\vec{K}_4 \cdot \vec{Q}_6$	$\vec{K}_4 \cdot \vec{Q}_7$	$\vec{K}_4 \cdot \vec{Q}_8$	
$\boxed{\text{roamed}} \rightarrow \vec{E}_5 \xrightarrow{W_k} \vec{K}_5$	$\vec{K}_5 \cdot \vec{Q}_1$	$\vec{K}_5 \cdot \vec{Q}_2$	$\vec{K}_5 \cdot \vec{Q}_3$	$\vec{K}_5 \cdot \vec{Q}_4$	$\vec{K}_5 \cdot \vec{Q}_5$	$\vec{K}_5 \cdot \vec{Q}_6$	$\vec{K}_5 \cdot \vec{Q}_7$	$\vec{K}_5 \cdot \vec{Q}_8$	
$\boxed{\text{the}} \rightarrow \vec{E}_6 \xrightarrow{W_k} \vec{K}_6$	$\vec{K}_6 \cdot \vec{Q}_1$	$\vec{K}_6 \cdot \vec{Q}_2$	$\vec{K}_6 \cdot \vec{Q}_3$	$\vec{K}_6 \cdot \vec{Q}_4$	$\vec{K}_6 \cdot \vec{Q}_5$	$\vec{K}_6 \cdot \vec{Q}_6$	$\vec{K}_6 \cdot \vec{Q}_7$	$\vec{K}_6 \cdot \vec{Q}_8$	
$\boxed{\text{verdant}} \rightarrow \vec{E}_7 \xrightarrow{W_k} \vec{K}_7$	$\vec{K}_7 \cdot \vec{Q}_1$	$\vec{K}_7 \cdot \vec{Q}_2$	$\vec{K}_7 \cdot \vec{Q}_3$	$\vec{K}_7 \cdot \vec{Q}_4$	$\vec{K}_7 \cdot \vec{Q}_5$	$\vec{K}_7 \cdot \vec{Q}_6$	$\vec{K}_7 \cdot \vec{Q}_7$	$\vec{K}_7 \cdot \vec{Q}_8$	

“Attend to”

	a	fluffy	blue	creature	roamed	the
$\vec{E}_1$	$\vec{E}_1$	$\vec{E}_2$	$\vec{E}_3$	$\vec{E}_4$	$\vec{E}_5$	$\vec{E}_6$
$\vec{Q}_1$	$\vec{Q}_1$	$\vec{Q}_2$	$\vec{Q}_3$	$\vec{Q}_4$	$\vec{Q}_5$	$\vec{Q}_6$
a	$\vec{E}_1 \xrightarrow{w_k} \vec{K}_1$	•	•	•	•	•
fluffy	$\vec{E}_2 \xrightarrow{w_k} \vec{K}_2$	•	•	+93.0	•	•
blue	$\vec{E}_3 \xrightarrow{w_k} \vec{K}_3$	•	•	+93.4	•	•
creature	$\vec{E}_4 \xrightarrow{w_k} \vec{K}_4$	•	•	•	•	•



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

	Q_1	Q_2	Q_3	Q_4	Q_5	$\dots$	Q_n	
K_1	$\frac{Q_1 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_1}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_1}{\sqrt{d_k}}$	
K_2	$\frac{Q_1 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_2}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_2}{\sqrt{d_k}}$	
K_3	$\frac{Q_1 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_3}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_3}{\sqrt{d_k}}$	
K_4	$\frac{Q_1 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_4}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_4}{\sqrt{d_k}}$	
K_5	$\frac{Q_1 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_5}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_5}{\sqrt{d_k}}$	

Unnormalized Attention Pattern

+3.53	+0.80	+1.96	+4.48	+3.74	-1.95
$-\infty$	-0.30	-0.21	+0.82	+0.29	+2.91
$-\infty$	$-\infty$	+0.89	+0.67	+2.99	-0.41
$-\infty$	$-\infty$	$-\infty$	+1.31	+1.73	-1.48
$-\infty$	$-\infty$	$-\infty$	$-\infty$	+3.07	+2.94
$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	+0.31

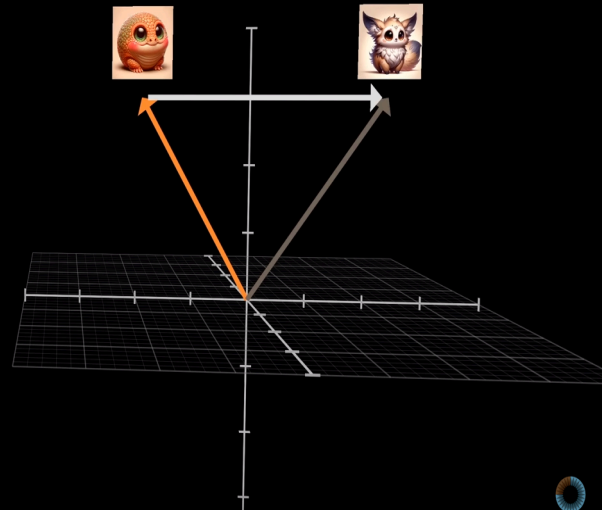
softmax
→

Normalized Attention Pattern

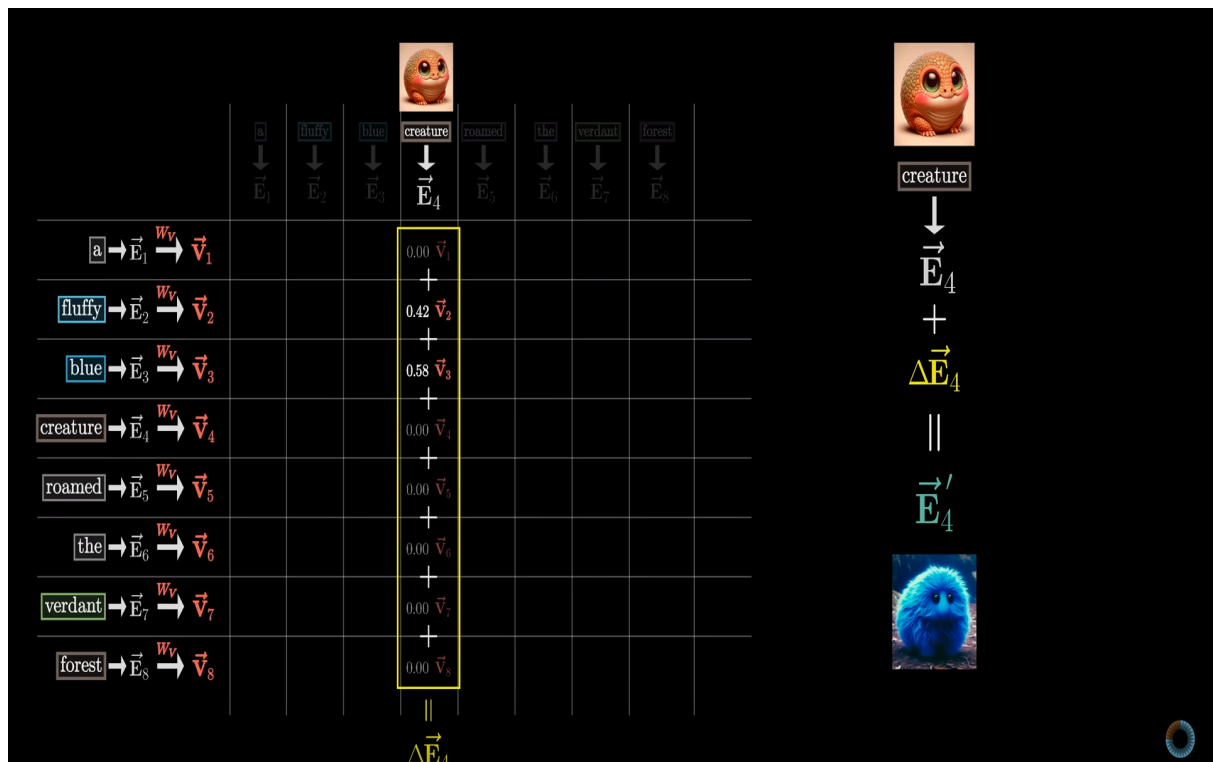
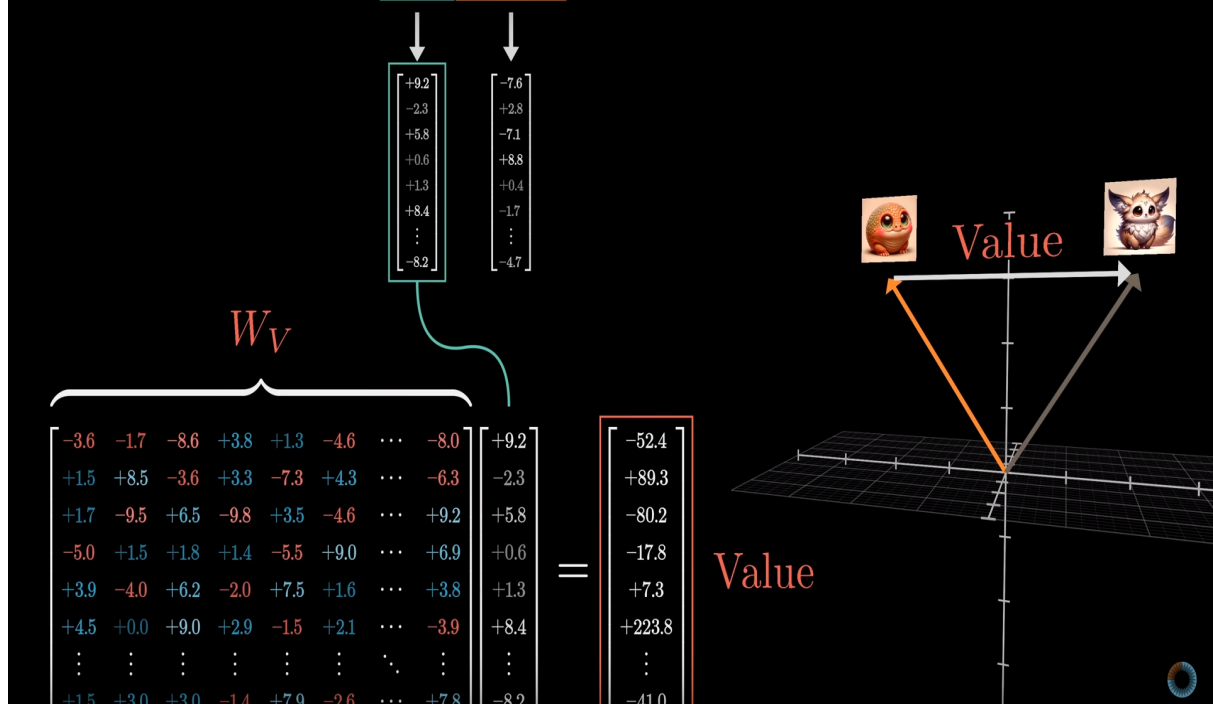
1.00	0.75	0.69	0.92	0.46	0.00
0.00	0.25	0.08	0.02	0.01	0.46
0.00	0.00	0.24	0.02	0.22	0.02
0.00	0.00	0.00	0.04	0.06	0.01
0.00	0.00	0.00	0.00	0.24	0.48
0.00	0.00	0.00	0.00	0.00	0.03

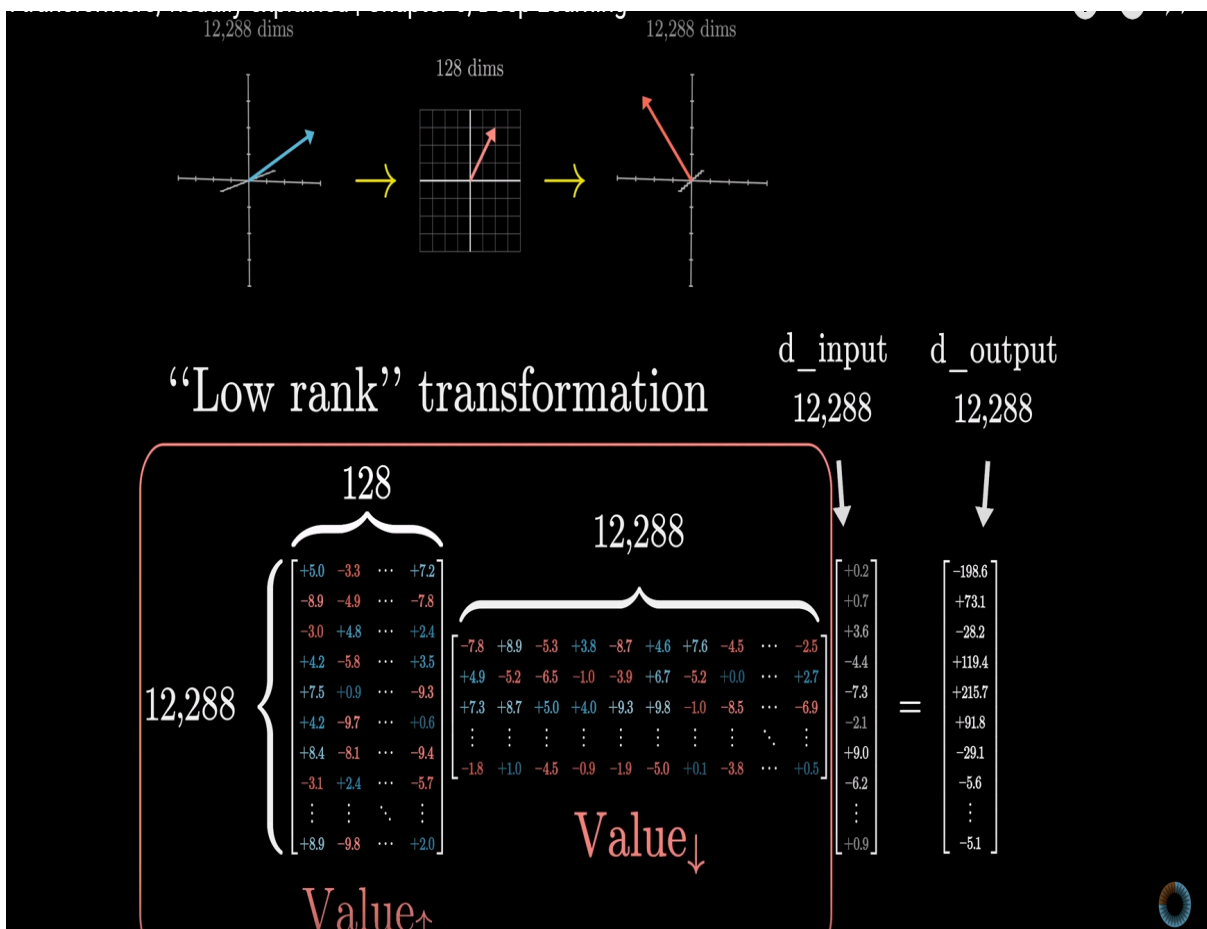
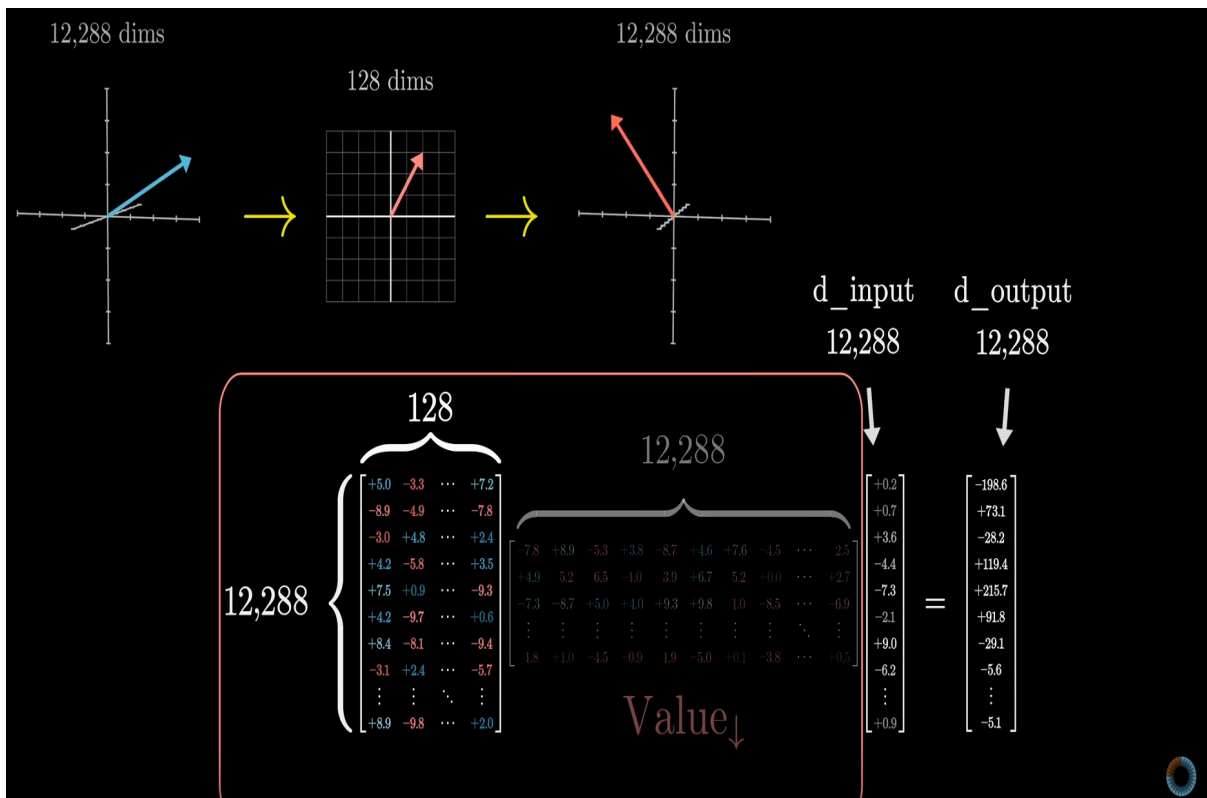
fluffy creature

↓	↓
$\begin{bmatrix} +9.2 \\ -2.3 \\ +5.8 \\ +0.6 \\ +1.3 \\ +8.4 \\ \vdots \\ -8.2 \end{bmatrix}$	$\begin{bmatrix} -7.6 \\ +2.8 \\ -7.1 \\ +8.8 \\ +0.4 \\ -1.7 \\ \vdots \\ -4.7 \end{bmatrix}$

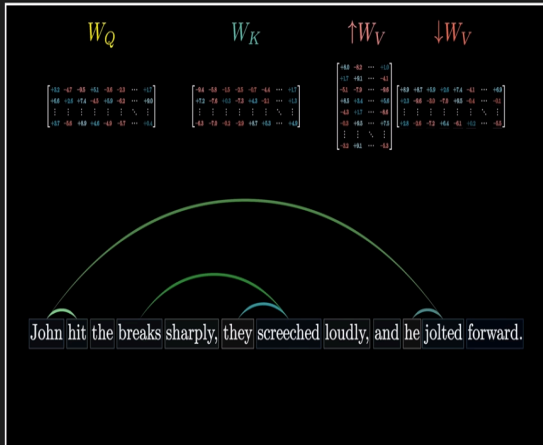


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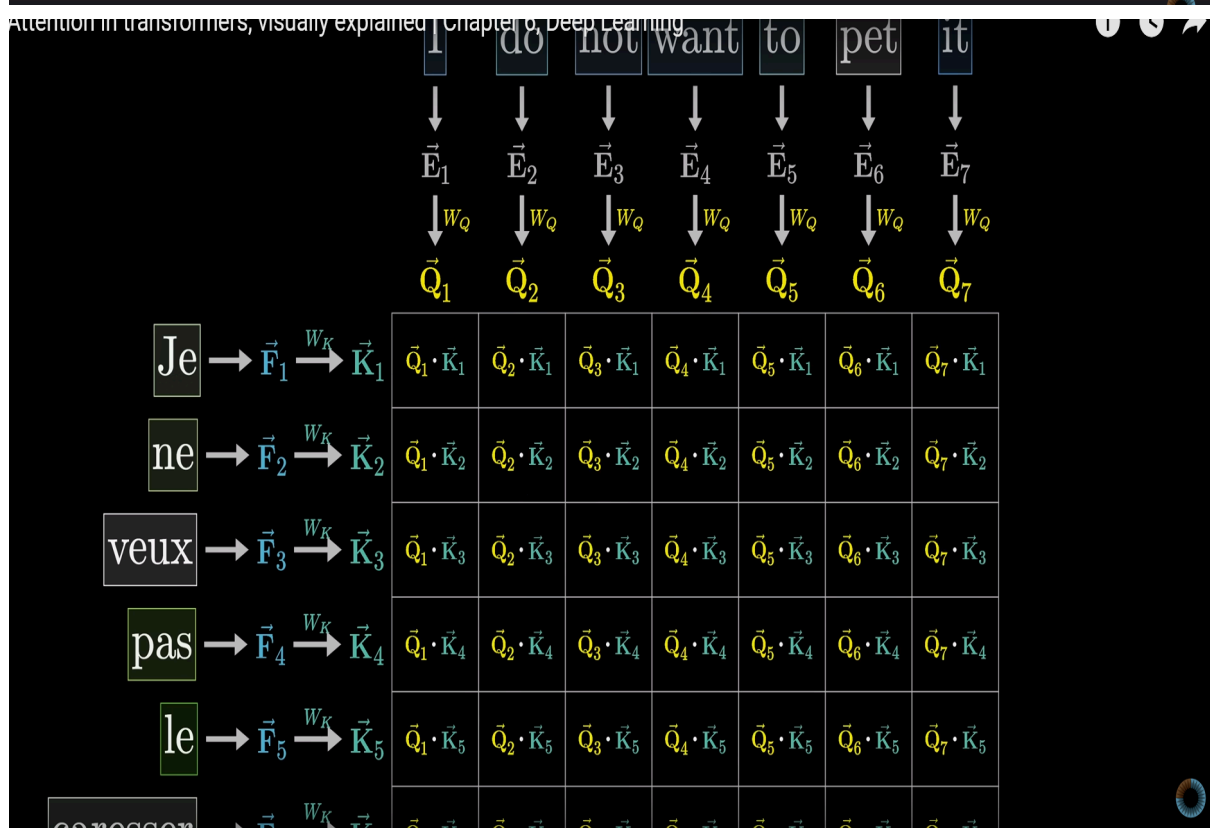
Self-attention

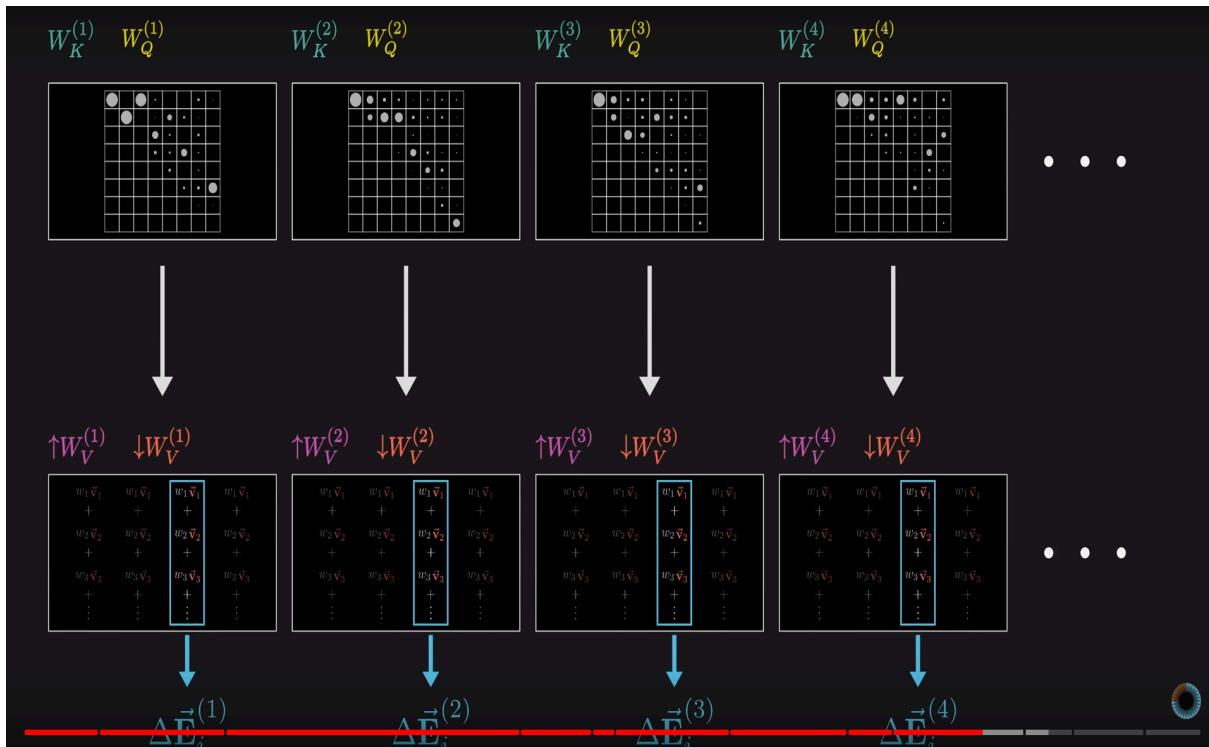


Cross-attention

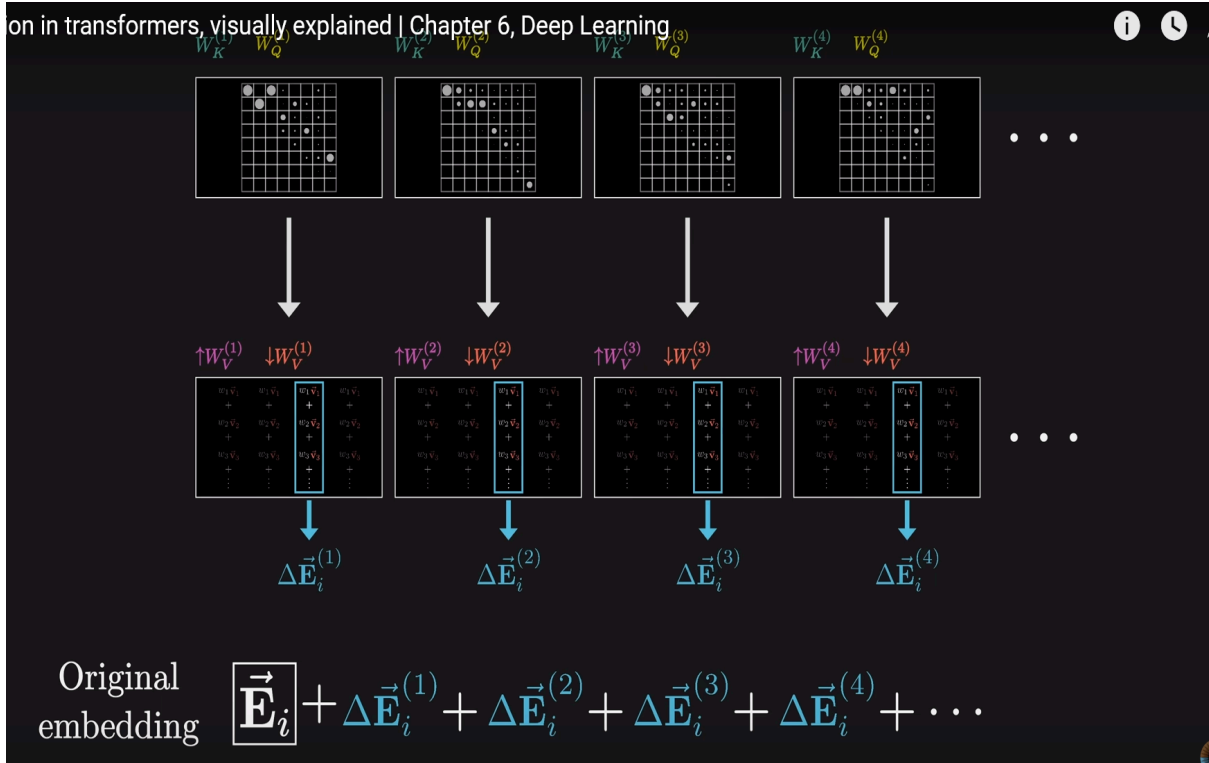


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Attention

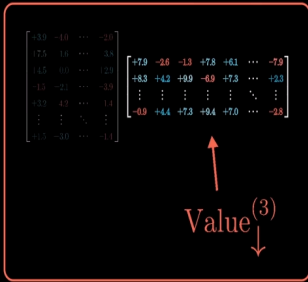
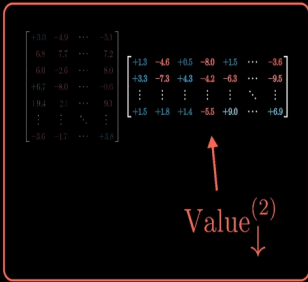
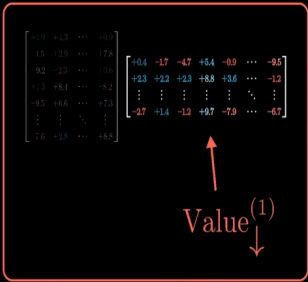


New
embedding

$$\vec{\mathbf{E}}_i + \Delta\vec{\mathbf{E}}_i^{(1)} + \Delta\vec{\mathbf{E}}_i^{(2)} + \Delta\vec{\mathbf{E}}_i^{(3)} + \Delta\vec{\mathbf{E}}_i^{(4)} + \dots$$

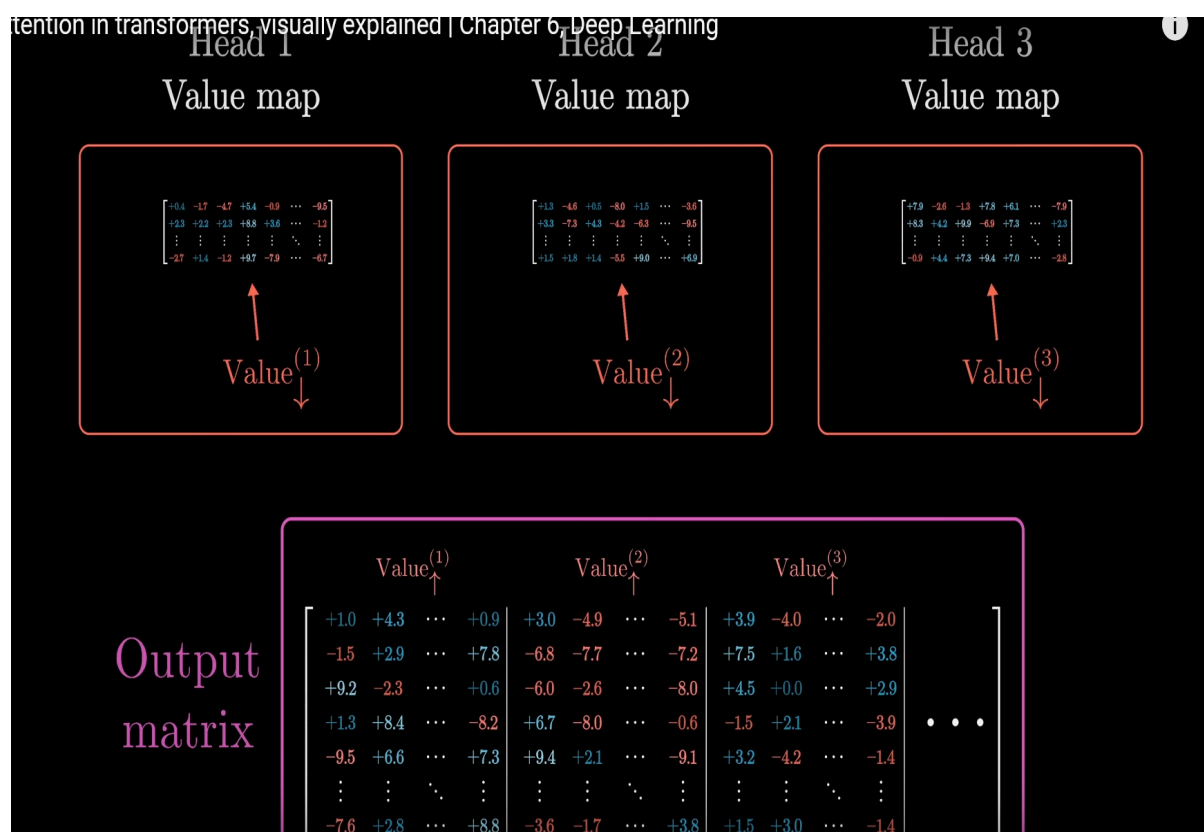
Head 1 Head 2 Head 3

Value map Value map Value map



Output
matrix





As presented so far, the value vectors inside an attention head would have the same dimension as the embedding space. This would involve taking weighted sums of these large vectors, weighted by the attention pattern.

You can save on computation by instead running the weighted sums on the smaller intermediate outputs produced by the value-down matrices.

That is, inside each head, use only the value-down matrix to produce a sequence of 128-dimensional vectors. You take weighted sums of these with the attention pattern, meaning the head outputs a single 128-dimensional vector for each position in the context.

You could then multiply each of those by the head's value-up matrix to get a 12,288-dimensional vector that can be added to the embedding in that position.

However! All those up-projections are usually bundled together (more confusion!). For each position in the context, you concatenate all the 128-dimensional vectors produced by each head at that position, creating a big vector with length (96*128). Now multiply this by the output matrix described on the previous slide.

This is mathematically equivalent to applying all the up-projections and adding the results (exercise). Given the chance, ML people love to compress things into a single matrix multiplication, even if it comes at the cost of conceptual clarity.